



# Dhruv Bhardwaj

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## Education

University of British Columbia

B.Sc. in Computer Science

*Deep Learning Specialization – DeepLearning.AI*

## Experience

Software Engineer – Visualization & Emerging Media Studio (UBC)

*Sept 2024 – Present*

- **Built real-time distributed backend systems** supporting multi-user scientific visualization across heterogeneous environments.
- **Optimized latency-sensitive data pipelines** (<80 ms) integrating WebSocket services and structured input streams for reliable interaction.
- **Diagnosed and resolved production-level issues** across networking, synchronization, and data consistency layers.

Backend Lead – Exam Generation & Analysis System (UBC Capstone)

- **Architected a microservice-based backend platform** using Node.js, TypeScript, tRPC, Prisma, and PostgreSQL.
- **Designed data pipelines** processing OMR submissions for **1000+ students per exam cycle**, enabling automated grading and analytics.
- **Implemented deterministic grading system** with snapshot-based versioning for reproducibility and auditability.
- **Enforced multi-tenant security** using role-based access control and PostgreSQL Row-Level Security (RLS).

Software Engineer – GitHub Analytics (UBC)

- **Built Python-based data pipelines** to analyze large-scale GitHub activity (commits, PRs, reviews).
- **Developed anomaly detection heuristics** and SQL workflows to improve signal extraction from noisy datasets.
- **Refactored legacy modules and improved test coverage**, increasing system reliability and maintainability.

## Selected Projects

PillCare – Multi-Modal Pill Identification System

*TensorFlow · React Native · Node.js · OpenFDA API*

- **Built a multi-stage ML pipeline** combining vision (MobileNetV2) and CRNN-based OCR for real-world pill identification.
- **Designed gated modality fusion** to dynamically weight visual vs text signals, improving robustness under ambiguous inputs.
- **Developed backend APIs** for real-time drug metadata retrieval and cross-platform deployment.

Companion – LLM-Powered Support System

*TypeScript · Node.js · LLM APIs · React Native*

- **Designed a multi-stage LLM request pipeline** with structured routing, safety classification, and deterministic fallback logic.
- **Implemented risk-aware response handling** (crisis detection + controlled bypass) to ensure safe and reliable outputs.
- **Built backend systems** for conversation state management, logging, and auditability of AI interactions.

Distributed Load Testing & Performance Benchmarking

*TypeScript · Node.js · Docker · CI/CD*

- **Building a distributed load-testing framework** to identify bottlenecks and integrate performance regression checks into CI pipelines.

## Technical Skills

**Languages:** Java, Python, TypeScript, C#, SQL

**Backend & APIs:** Node.js, REST APIs, tRPC, Prisma, FastAPI, WebSocket

**Systems:** Distributed Systems, Data Pipelines, Microservices

**Data:** SQL, Relational Modeling, ETL, Query Optimization, PostgreSQL

**Tools:** Docker, CI/CD, Linux, Git